Cryptography in Cybersecurity

UNIT-II : Assymmetric Key Encryption.
Principles of public key Components.
The RSA algorithm - key Management
Diffie Hellman key exchange
Elliptic curve arithmetic
Elliptic curve cryptography.

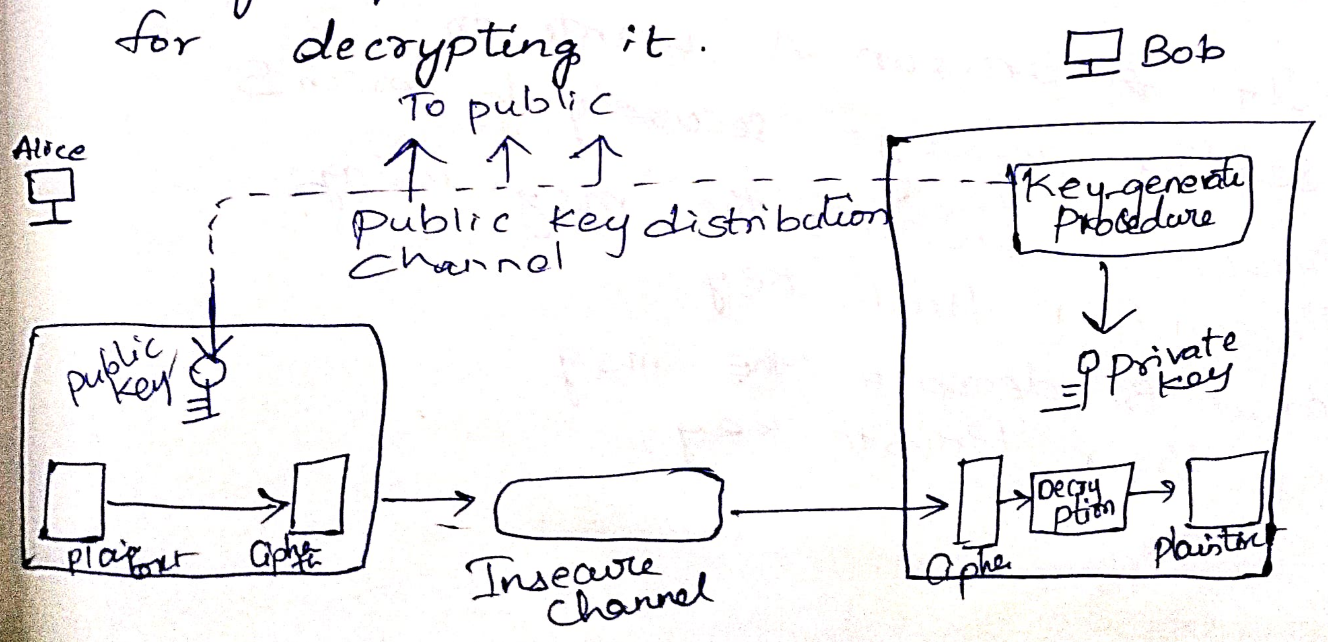
ASSYMMETRIC KEY ENCRYPTION:

→ In asymmetric key cryptography, → There are 2 keys (key pairs) → 1. public key
2. Private key

" The public key is publicly distributed. Anyone can use this public key to encrypt messages, but only the recaipient, who holds the corresponding private key, can decrypt those messages.".

Challenges: Key distribution
Digital Signatures.

→ Assymetric ~~key~~ algorithm uses one key for encrypting data and another, related key for decrypting it.



Principles of public key cryptosystems:

→ Confidentiality → Through need of key distribution, where users seeking private connection exchange encryption keys.

RSA Algorithm:

(Rivest - Shamir - Adleman) is an asymmetric or public-key cryptography algorithm.

→ works on 2 different keys.

1. Public key → used for encryption and is known to everyone.
2. Private key → used for decryption and must be kept secret by the receiver.

- published in 1977.

Eg: If ~~A~~ person A wants to send a message securely to person B

- Person A encrypts the msg using Person B's Public key
- Person B decrypts the msg using their Private key.

3 steps:

1. Key Generation
2. Encryption
3. Decryption

1. Key Generation:

→ Select 2 large prime numbers p and q
→ calculate $n = p * q$
→ Calculate $\varphi(n) = (p-1)(q-1)$ where φ is Euler Totient function
→ Chose an integer e ∋ $1 < e < \varphi(n)$ and $\gcd(e, \varphi(n)) = 1$
→ Compute d, the modular multiplicative inverse of e modulo $\varphi(n)$
→ Public key: (e, n)
→ Private key: (d, n)

2. Encryption:

→ Convert plaintext message into an integer m
→ Compute ciphertext $c = m^e \bmod n$

3. Decryption:

Calculate plaintext message
$m = c^d \bmod n$,

Example for LSA Algorithm :

1. Key Generation (Performed by the receiver)

Receiver first generates a public and private key pair using following steps:

→ choose 2 distinct prime numbers, p and q.

let $p = 3$, $q = 11$

→ Compute n, the modulus.

$n = p \times q = 3 \times 11 = 33$.

↳ This n will become part of both public and private keys.

→ Compute Euler's totient function $\phi(n)$

$\phi(n) = (p-1)(q-1)$
$= (3-1)(11-1)$
$= 2 \times 10$
$= 20$.

→ chose a public exponent e. $(1 < e < \phi(n))$

e → must be coprime to $\phi(n)$.

let's choose $e = 7$ [since $(GCD(7,20) = 1$]

→ Compute the private exponent d.

→ d must statisfy modular multiplicative inverse condition

$(d \times e)(\text{mod } \phi(n)) = 1$.

→ we need to find d →

$(d \times 7)(\text{mod } 20) = 1$

⇒ $d = 3$ (∵ $(3 \times 7)(\text{mod } 20) = 21 \text{ mod } 20 = 1$

* Final keys: public key (shared with anyone)
$(e, n) = (7, 33)$

Private key (kept secret by the receiver) : $(d, n) = (3, 33)$

2. Encryption (Performed by the sender)

Suppose sender wants to send a msg m, which they convert into a number (eg ASCII values).

→ Let the message number be $m = 2$

The sender uses recipients public key (e, n) to calculate the cipher text c,

using: $c = m^e \pmod{n}$

$\Rightarrow c = 2^7 \pmod{33}$

$\Rightarrow c = 128 \pmod{33}$

$\Rightarrow \frac{128}{33}$ → gives remainder 29

$\Rightarrow$ The Ciphertext c is 29

3. Decryption: (Performed by the receiver)

The receiver receives the Ciphertext $c = 29$.

→ They use their private key (d, n) to recover the original msg m.

using:

$m = c^d \pmod{n}$

$\Rightarrow m = 29^3 \pmod{33}$

$\Rightarrow m = 24389 \pmod{33}$

$\Rightarrow 24389 \div 33$ → gives remainder 2.

∴ The original msg ~~is~~ m is 2,

∴ The algorithm successfully recovers the original message.